

Flash Gordon Documentation

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Single-molecule Imaging Center (SMC)

St Jude Children's Research Hospital

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WARNING: by installing this software, the user assumes all risks and liability for any potential injury. The software is intended to be utilized by experienced professional academic scientists in accordance with all applicable laser safety and other safety procedures to minimize the potential risk of harm to users by high-power light sources. **The software does not provide any safety measures**, which must be implemented separately and fully validated by the user prior to operation.

Introduction

This software is intended for the acquisition of image stacks (movies) for single molecule imaging experiments utilizing total internal reflection fluorescence (TIRF) microscopes and wide-field sCMOS cameras. It is designed to be used with the SPARTAN data analysis software.

Please cite the following paper if you use Flash Gordon in your laboratory:

Juette MF, Terry DS, Wasserman MR, Altman RB, Zhou Z, Zhao H, Blanchard SC. Single-molecule imaging of non-equilibrium molecular ensembles on the millisecond timescale. *Nature Methods*. 2016 Apr;13(4):341-4. doi: 10.1038/nmeth.3769. Epub 2016 Feb 15.

To be updated with our manuscript in preparation here

This software is intended only for research use in an academic setting. For licensing information, please see “[license.txt](#)” file associated with the software.

If you have comments, concerns, bug reports, or feature requests, please contact Scott Blanchard (scott.blanchard@stjude.org).

Limitations

This software is intended only for the devices listed below and is not designed to support all possible workflows. Below is a list of specific limitations the user should be aware of. To add support for new devices, please see the section at the end of this document titled “Adding support for new hardware devices”.

- The software is designed to work with an instrument in which every camera is dedicated to a specific fluorophore spectral band. If you use the software with an optical device to project multiple spectral images onto a single camera, most functionality will perform correctly, but the metadata in saved movie files will not be correct.
- Flash Gordon uses an external device (DAQ) to trigger each camera frame. This approach has a maximum frame rate of 500 s^{-1} (2 ms frame intervals). For faster imaging rates, only the first frame is triggered by the external device and all cameras run on their own internal clocks. This approach may result in a loss of synchronization over long time periods (seconds to minutes, depending on how well matched are the camera internal clocks), but that this is not generally a problem given the fast photobleaching of commonly used fluorophores under the intense illumination required for such short exposure times. Especially with short exposure times, we advise utilizing an oscilloscope to monitor device synchronization.
- All devices the user intends to use must be connected and properly initialized before starting Flash Gordon. Disconnecting a device while Flash Gordon is running may result in crashes or unexpected behavior. All devices listed in the configuration file must be detected or Flash Gordon will not start.
- Flash Gordon assumes cameras use 2x2 pixel binning. All cameras in use at the same time **must** have the same model number.

Recommended Hardware

- Up to four Hamamatsu Fusion cameras (C14440-20UP). For optimal results, these should have sequential serial numbers and the same firmware version.
<https://www.hamamatsu.com/eu/en/product/cameras/cmos-cameras/C14440-20UP.html>
- Each camera is dedicated to a specific fluorophore's spectra band using an optical splitter such as Cairn MultiCam with appropriate dichroics and band-pass filters. The system should be adjusted so the camera fields of view are closely aligned and parfocal.
<https://www.cairn-research.co.uk/product/multicam/>
- Nikon Eclipse Ti2 microscope stand with motorized filter turret, stage, and Z-drive objective mount.
- Four Uniblitz LS6 shutters (for 473, 532, 640, 721 nm laser lines) with VCM-D4 driver:
<https://www.uniblitz.com/products/ls6/>
<https://www.uniblitz.com/products/vmm-d4-shutter-driver/>
- Optional: Thorlabs PRM1Z8 for rapid laser power attenuation with a rotating polarizer.
<https://www.thorlabs.com/thorproduct.cfm?partnumber=PRM1Z8>
- National Instruments USB-6501 DAQ for triggering shutters and cameras.
- Dell Precision 7920 or similar with dual Xeon processors, 128 GB RAM, and Intel Optane 905P PCIe SSD (1.5 TB). If another SSD is used, make sure it has a measured sustained write speed of at least 1 GB/s (gigabytes not gigabits).

Supported Hardware

- **Cameras:** Hamamatsu Flash 4.0 v2 (C11440-22CU), Flash 4.0 v3 (C13440-20CU), Fusion (C14440-20UP), and Fusion-BT (C15440-20UP). Hamamatsu Quest (C15550-20UP) has preliminary support but is untested.
 - For optimal results, follow all manufacturer recommendations for hardware installation, including the use of one PCIe USB card per camera, and software configuration, such as BIOS settings.
- **Microscope stand** (filter and light path selection)
 - Nikon Eclipse Ti2 (USB connection)
- **Stage controller**
 - Nikon Eclipse Ti2 (USB connection)
 - Newport ESP301 (tested with LTA-HS motors). Must be connected by USB.

- ASI controllers (tested with MS-2000 stage)
Must be connected by serial port (RS232) with 9600 baud rate.
- Ludl MAC5000 or MAC6000 controller (tested with BioPrecision2 stage).
Must be connected by serial port (RS232) with 9600 baud rate.
- **DAQ** (for triggering shutters and cameras)
 - National Instruments USB-6501. Many NI DAQ devices are likely supported. Only the port and channel addresses would need to be changed.
- **Laser power control**
 - LaserQuantum Opus, Ventus, and Gem lasers via the RS232 interface. Other LaserQuantum lasers should work but are untested.
 - Coherent OBIS lasers implemented but untested.
 - Thorlabs PRM1Z8 motorized rotation stage (for laser power attenuation).

Hardware Setup Guide

- For a general overview of the experimental setup, please see the following paper:
<https://www.nature.com/articles/nmeth.3769>
- Configure the shutter controller(s) so that the shutters are closed with 0V input and open with +5V input. With Uniblitz drivers, connect the appropriate DAQ output ports to the “pulse input” port.
- The camera trigger DAQ line should be connected to the “external trigger” input of all Hamamatsu cameras. Optional: connect port #1 of the camera to an oscilloscope to monitor synchronization.
- All cameras being used together must have the same camera model. USB, CameraLink, and CoaXPress interfaces are supported, but the slower USB interface is assumed for timing calculations.
- Control software associated with each device, such as the RemoteApp program for LaserQuantum lasers can be useful both to verify the device is properly connected and determine the COM port and other connection parameters for it.
- Follow all manufacturer recommendations for camera hardware installation, including the use of one PCIe USB card per camera, appropriate device drivers, software configuration, and BIOS settings.
- Recommended: disable power management in Device Manager for all USB devices (and any hubs they depend on). Otherwise, devices may randomly fail when put to sleep.
- If your instrument includes the Thorlabs PRM1Z8 device, whenever this device is turned on, you must manually set the home position. To do so, click the MENU button, scroll to

“Start Homing”, and press MENU button again. This process is not performed automatically by Flash Gordon.

Installation (stand-alone executable version)

The stand-alone executable version is recommended for all users unless you plan to customize the code to add new devices. Before installing Flash Gordon, please install the following pre-requisites:

- 64-bit LabVIEW run-time engine 2018 (SP1):
<https://www.ni.com/en-us/support/downloads/software-products/download.labview-runtime.html#309628>
- DAQ-mx version 21.8 (for the USB-6501 device):
<https://www.ni.com/en-us/support/downloads/drivers/download.ni-daqmx.html#445931>
- Latest version of DCAM (for Hamamatsu cameras). We have tested most extensively with versions 22.2.6391 and 22.8.6486. <https://dcam-api.com/>
- Nikon Ti2 SDK 64-bit, which includes the device driver for the Ti2 microscope.
<https://nisdk.recollective.com/microscopes>
- Optional: Thorlabs Kinesis (for Thorlabs PRM1Z8).
https://www.thorlabs.com/software_pages/ViewSoftwarePage.cfm?Code=Motion_Control

Once these packages are installed, unzip FlashGordon.zip to “C:\FlashGordon” and create a desktop shortcut to FlashGordon.exe. Ensure all devices are initialized and connected. Run FlashGordon.exe to start the program.

Installation (source code version)

The source code version is only recommended for advanced users experienced with LabVIEW programming. Before installing Flash Gordon source code, please install the following pre-requisites:

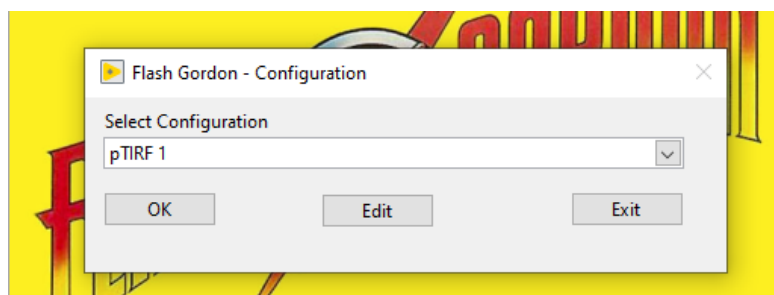
- 64-bit LabVIEW Professional version 2018 SP1. This requires a license to be purchased.
<https://www.ni.com/en-us/shop/labview.html>
- VI Package Manager (VIPM) and from there install “UI Control Suite: System Controls 2.0”. <https://www.ni.com/en-us/support/downloads/tools-network/download.jki-vi-package-manager.html#443251>
- DAQ-mx version 21.8 (for the USB-6501 device):
<https://www.ni.com/en-us/support/downloads/drivers/download.ni-daqmx.html#445931>

- Latest version of DCAM. We have tested most extensively with versions 22.2.6391 and 22.8.6486. <https://dcam-api.com/>
- Install Hamamatsu Video Capture library for LabVIEW (tested with version 4.4.23): <https://dcam-api.com/hamamatsu-software/>
- Nikon Ti2 SDK 64-bit, which includes the device driver for the Ti2 microscope. <https://nisdk.recollective.com/microscopes>
- Optional: Thorlabs Kinesis SDK for LabVIEW https://www.thorlabs.com/software_pages/ViewSoftwarePage.cfm?Code=Motion_Control
- Optional: ASI stage controller driver
This is included in the “Hardware/Stage Controller/driver vi/ASI TG-1000.zip” directory. Unzip this file into “C:\Program Files\National Instruments\LabVIEW 2018\instr.lib\”. <https://www.asiimaging.com/support/downloads/ms-2000-control-using-nis-labview-and-the-serial-port/>

Once these packages are properly installed, unzip the Flash Gordon source code to “C:\FlashGordon”. To run Flash Gordon, open the project file, double click on “Flash Gordon.vi” from within the LabVIEW project, and click the “play” button to start the program. **Do not start “Flash Gordon.vi” from outside the project!**

Configuration

Flash Gordon is designed to support a variety of hardware configurations, with the settings for each instrument contained within an XML file in the “Config” folder. When Flash Gordon starts, it will prompt the user to select amongst these files to determine which devices are present and the settings to use. You can modify any existing configuration files by clicking the “Edit” button below the list box. Clicking “Ok” on this dialog will overwrite the selected configuration file. To remove a configuration, simply delete the corresponding file in the Config folder. To rename a configuration, simply rename that XML file. Finally, to make a new configuration, you can copy an XML file with a new name. You must restart Flash Gordon for it to detect new or renamed files.



To create a new configuration file in the source code version, you have the additional open to use “Application\Create Hardware Configuration.vi” in the LabVIEW project. Press Control-E to open the wire diagram and modify the elements according to the devices installed on your instrument. Click run and save the resulting XML file with a name that is easily recognizable.

Below is a list of the configuration options:

- **Shutters**

- **IO Line:** this is the virtual device path provided by the DAQ that controls the shutter. You can use the NI MAX utility to determine the port names and test them. It should be connected to the “pulse input” line of the shutter driver.
- **Name:** name of the camera channel associated with this laser.
- **Wavelength:** wavelength of the laser line in nanometers.

- **Stage Configuration**

- **Stage Type:** “ESP301” and “Ti2” are supported; choose “Disabled” if you do not have a supported stage installed.
- **Serial Port:** only needed for Newport ESP301 stage controller.
- **Forward Step** (mm): sets how far the stage moves between successive movies (X direction). Controls the “Move Forward” button in the Flash Gordon front panel. Should be adjusted so that the illumination intensity drops to zero at this distance so that the next field is not photobleached by the previous movie.
- **Lane separation** (mm): controls how far the stage moves between successive lanes (Y direction).
- **Lane Length** (mm): controls the usable area of the channel in the “forward step” direction. Adjust this to control the default photobleaching length. **WARNING: be careful to check that this value does not position the prism to make contact with fluidics or other obstructions that could damage it.**

- **Microscope Configuration**

- **Microscope Type:** Specifies stage controller type. Set to “Disable” if you do not have a supported microscope stand installed. The options “Eclipse Ti” and “TE2000” are not supported.
- **Serial port** (for stages connected via RS232)
- **Detection settings:** this populates the “Detection” dropdown list to control the position of the filter block and light path inside the microscope. For the Ti2 microscope, the light path values are as follows:
 1. Eyepieces
 2. Right port
 3. Bottom (U) port

4. Left port

- **Camera Driver configuration**

- **Sync line:** virtual DAQ port connected in parallel to each camera “Trigger Input” port. This is used to trigger camera exposure to ensure synchronized.

- **Laser configuration**

- **Laser Type:** “LaserQuantum” and “OBIS” will allow the power at the laser head to be adjusted via RS232. “PRM-Z1” allows for controlling a motorized stage for laser power attenuation using a polarizer.
- **Serial Port:** for LaserQuantum and OBIS only.
- **Wavelength:** wavelength in nanometers of the laser line. Should be identical to the number listed in the corresponding shutter configuration or it will not be accessible in the user interface.
- **Polarizer theta zero** (PRM-Z1 only): polarizer angle in degrees where light transmission is highest.
- **Polarizer amplitude** (PRM-Z1 only): power in milliwatts of laser intensity reaching the sample when the polarizer is set to the angle above.

- **Installed Cameras**

- **Channel Name:** short string used to uniquely identify each camera. This should be one of the following: Cy2, Cy3, Cy5, Cy7. This field is not intended to specify the fluorophore or sample, but instead is the name of the spectral channel. This is used both in the user interface to identify each camera and in the metadata for automating analysis.
- **Serial Number:** see camera documentation and stickers on the camera. This is typically a six-digit number that uniquely identifies the camera.
- **Flip Vertically:** if true, raw images from this camera are flipped vertically so that it matches the orientation of the other cameras. If possible, arrange your optical system to avoid using any “flip” settings. Also please be aware that flipping vertically with sCMOS cameras may result in slight asynchrony between channels due to the line-by-line readout process; this will not be a problem in ALEX or stroboscopic modes and should not be an issue at all with Flash cameras.
- **Flip Horizontally:** if true, raw camera images from this camera will be flipped horizontally for display and saving to file so that it matches the orientation of the other cameras.
- **photonsPerCount** (optional): number of photoelectrons per each arbitrary unit of the camera digitizer, as specified in the data sheet from the camera manufacturer. This is only used for metadata to assist in automating analysis.

- **Wavelength:** wavelength in nanometers of the laser that is used to directly excite fluorophores in this spectral band. **For integration with SPARTAN, it is important that this exactly match the shutter/laser wavelength configured above.**

The camera list can have any number of entries associated with all cameras in a laboratory or just a subset for each instrument. The lists for all other devices should include those actually connected to the instrument.

The default configuration file can be specified by editing the Init.xml file. Change the “Hardware Configuration File” value to the name of the configuration file you want to be the default.

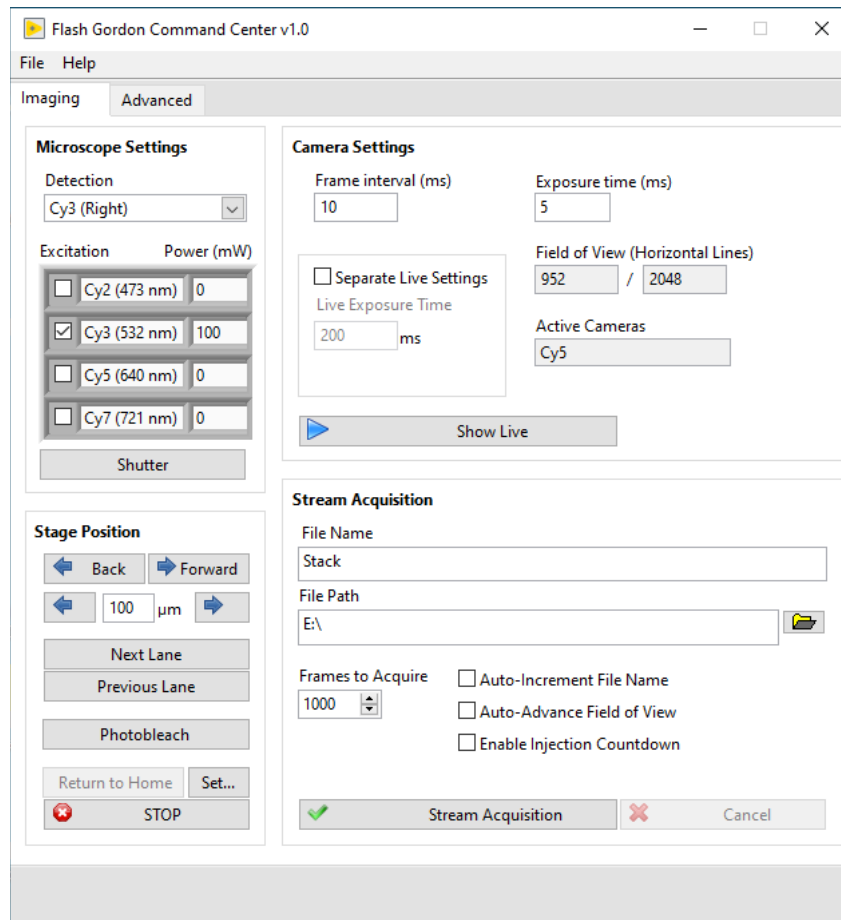
Be sure to verify synchronization. Connect the “trigger output 1” port of each camera to an oscilloscope and verify that the rising edges are synchronized within ~1 ms. Check this at varying exposure times between 200 ms and 2 ms. At times faster than 2 ms, the cameras are internally triggered and may lose synchronization over time; this is normal (see Limitations).

Using Flash Gordon

Startup

Turn on all devices, wait for them to finish their startup routine, and launch FlashGordon.exe (or Flash Gordon.vi for source code version). A dialog will appear asking which configuration to use. Select the name corresponding to the configuration XML file you created in the previous section and click “OK”. If an error dialog appears that says “No cameras detected”, verify that the cameras are properly connected, lights on the cameras have stopped blinking (startup completed) and that the serial numbers in the configuration file exactly match those on the devices. If the configuration dialog never appears, check that the “Init.xml” file is present and properly formatted.

Flash Gordon will attempt to connect to all configured devices. If successful, the Main Window will appear. If you receive any errors at this stage, close Flash Gordon and LabVIEW completely. (Open Task Manager and select End Task to kill the program if it does not shut down completely.) Verify that the configuration file is correct and that all devices are properly connected. If this does not work, try disabling any unnecessary devices to determine which is causing the problem.



The Main Window is used for controlling all devices and collecting data.

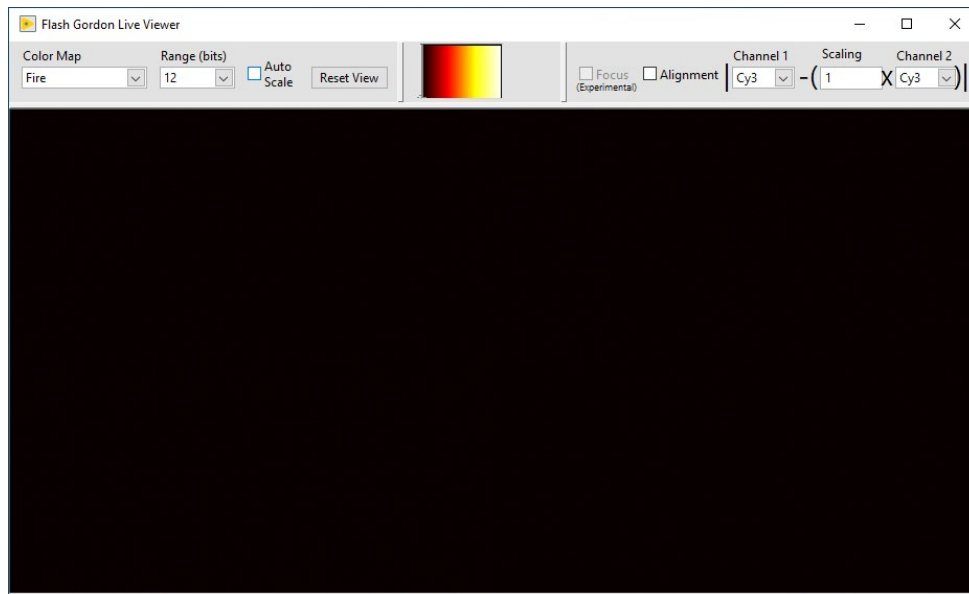
- **Detection:** specifies the microscope light path, including the filter set and imaging port.
WARNING: the “Lasers to Eyes” setting will direct unfiltered emission light directly to the eyepieces and could result in injury without proper care. This setting is intended to be used only for rough focusing at the beginning of an experiment with laser powers less than 5 mW.
- **Excitation:** the check boxes specify which laser line is active, which means it will be turned on in Stream Acquisition or Live modes or when the user manually clicks the “Shutter” button.
- **Power (mW):** change this value to adjust the laser power for this line (if configured). If the power is changed externally to Flash Gordon, the user interface will not be updated.
- **Shutter:** pressing this button will open/close the currently active shutter(s).
- **Exposure time (ms):** time that the camera is exposing and the laser shutters are open within each frame interval. In continuous illumination, this is the same as frame interval.
- **Frame interval:** time period between each frame. In continuous illumination modes, this is the same as the camera exposure time. To do time-lapse type recordings, you can simply set this value to a large number (e.g., 60 seconds).

- **Fields of View (Lines):** the first box shows the number of camera lines that will be used for stream acquisition. The second number is the maximum number of vertical lines supported by the camera sensor. The first number will change depending on the selected exposure time, ALEX mode, etc. Both numbers refer to physical camera lines, so the number of pixel rows in the saved movie files will be half that with 2x2 binning.
- **Detected Cameras:** lists the names of the cameras detected at startup. If some are missing, verify the cameras are properly connected and the serial numbers are correct in the configuration file before restarting Flash Gordon. For Hamamatsu cameras, the “DCAM Configurator” utility is useful for determining that cameras are indeed detected by the API; this can be accessed with the DCAM installed under “tools”.
- **Forward/Back:** moves the stage in the X direction by the interval specified in the configuration file to obtain a new field of view with unphotobleached molecules. This interval is also used clicking the “Stream Acquisition” button with “Auto-advance” checked.
- **Left/right arrows:** move stage in the X direction by the interval specified in the text box (in microns).
- **Next/Prev. Lane:** move the stage in the Y direction by the interval specified in the configuration file to obtain a new lane of fields.
- **Photobleach:** move the stage in the X direction in small increments for the full distance specified in the configuration file. The purpose of this mode is for the user to photobleach any unwanted molecules on the surface in the current lane. The user must manually adjust the laser power to maximum in order to have the desired effect.
- **Set (home):** use the current stage position as “home”. With the ESP301, this sets the zero position on the stage controller itself. With Ti2, this only sets a software zero position since the Ti2 does not have a configurable “home” position. Zeroing the position on the Ti2 control pad has no effect on this setting.
- **Return to home:** move stage to the zero position (if set). **Use care when setting the zero position to avoid damage to the stage and objective.**
- **STOP:** abort any current stage motion. Used primarily to interrupt photobleaching command.
- **Show Live:** opens the Live Viewer and opens any active shutter(s). Click again to close the viewer. If “Separate Live Settings” is checked, this value will be used for the camera frame interval instead of “Stream Exposure Time”. This option is intended to allow the user to position the stage and focus on the sample at low laser power (and correspondingly long exposure times) to avoid photobleaching the sample prematurely.
- **Stream Acquisition:** start acquiring a movie. Click “Cancel” to stop the acquisition early. You will be prompted to save the truncated movie if desired.
- **Stream Acquisition/File Name, File Path:** path to the movie file. Any file extension is ignored; it will always be .tif. Any folders must first be created by the user.

- **Frames to Acquire:** number of frames in a stream acquisition.
- **Auto-Increment File Name:** a three-digit number will be appended to the file name that is incremented with each acquisition so that the user doesn't need to keep changing the file name field.
- **Auto-Advance Field of View:** each time the user clicks "Stream Acquisition" the stage is automatically moved forward by the increment set in the configuration file, which is the same as clicking the "Forward" button.
- **Enable Injection Countdown:** opens a timer that counts down to zero for when the shutters open and the acquisition begins. This mode is useful for timing manual injections with the start of acquisition.

Live Viewer

To see acquired images from the cameras without saving any data, click the "Show Live" button in the main window. A new window will appear showing the frame data from all active cameras.



The Live Viewer window includes several controls at the top. These only affect the display of the pixel data and have no effect on the acquired movies.

- **Color Map:** changes the mapping of monochrome pixel intensity values to colors for easier viewing. (Source code version: new colors can be added by modifying the file "Resources\Helper VIs\Color Table To Reference.vi" and the corresponding enum "Resources\Type Defs\Color Map.ctl").
- **Histogram:** the colorful plot in the middle displays a histogram of pixel intensity values. The two vertical bars allow the user to change the mapping of pixel values onto the color scale (as long as "Auto Scale" is not selected). In most cases, the lower bar should

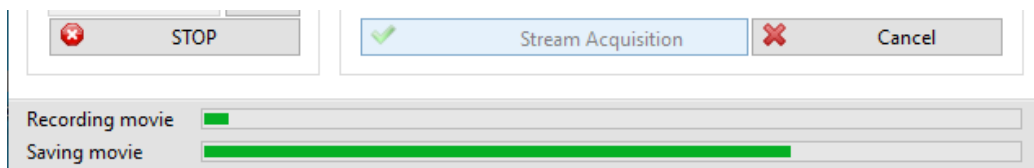
be set on the large peak corresponding to background pixels, while the upper bar is set so that the molecule pixel intensities are clearly visible but not over saturated, especially when adjusting the focus.

- **Range (bits):** This changes the pixel value range of the histogram. While the sensor may have a 16-bit dynamic range, the lower intensities in single molecule fluorescence experiments are typically better suited for a 13-bit range.
- **Auto Scale:** if checked, the histogram scale bars will be automatically adjusted to be in an optimal range.
- **Alignment:** if checked, the Live Viewer will display the image of one camera subtracted from another. This is useful when manually adjusting the alignment between cameras. The scaling parameter is useful when two spectral bands have very different intensities and is most often used with far-red dyes like Cy7.

NOTE: When looking at the Live Viewer images, you may notice that the cameras seem unsynchronized, that the refresh rate is significantly slower than the selected acquisition rate, or that the shutters open and the movie is acquired without any frame data displayed at all (for very short movies). This is all normal. Flash Gordon is a multi-threaded application where the Live Viewer executes in a separate thread from device management and only updates every ~100 ms. The correct way to verify that the devices are properly synchronized is to use an oscilloscope connected to the outport ports of the cameras and other devices.

Stream Acquisition

To acquire a movie, click the “Stream Acquisition” button. The Live Viewer window will appear and the cameras will stream data to disk (in the form of .dcimg files). A progress bar will appear in the bottom of the main window next to “Recording Movie” to show how many frames have been recorded. When the acquisition is complete, the live viewer will close. The raw data is converted into a .tif file in the background, with the progress displayed at the bottom of the main window next to “Saving Movie”. There is no problem with starting an acquisition while the previous movie is still saving. If you click “Cancel” or close the live viewer window, you will be prompted on whether to save the already-acquired frames.



Stroboscopic and time-lapse recordings

By default, movies are collected with continuous illumination, with the active shutter lines opening for the first frame and closing after the last. In some experiments, it may be desirable to open the shutter for only a short time during each frame. This approach, called “stroboscopic

illumination” allows the imaging time resolution to be controlled by the light source rather than the camera. This permits imaging with the full camera frame with exposure times as short as 2 ms, whereas with continuous illumination such short exposure times are only possible with a significantly reduced imaging field of view due to limitations in camera pixel readout rates. By contrast, the illumination period is only limited by the response time of the shutter.

Advanced Illumination Settings

☐ Alternating Laser Excitation (ALEX)

☒ Stroboscopic illumination / time-lapse

Duty Cycle: Max shutter rate (Hz)

To enable this mode, click on the “Advanced” tab and select “stroboscopic illumination”. The “duty cycle” field displays the fraction of each frame that is illuminated (shutter is open). Note that all shutters have a minimum exposure period. For the recommended LS6 shutters, this is ~2 ms, but will likely be longer for other shutters. Exposure periods less than recommended by the shutter manufacturer may not allow the shutter to open completely, resulting in uneven and diminished sample illumination, but should not harm the shutter. Check your shutter data sheet for details.

Alternating Laser Excitation (ALEX)

For some experiments, it is desirable to trigger changes in illumination wavelength timed with each camera frame. This approach, called alternating laser excitation or ALEX, can provide additional information including the stoichiometry of fluorophores and may also be useful for calculating FRET efficiencies in 3- and 4-color schemes.

Advanced Illumination Settings

☒ Alternating Laser Excitation (ALEX)

☐ Stroboscopic illumination / time-lapse

Duty Cycle: Max shutter rate (Hz)

To enable ALEX mode, click on the “Advanced” tab at the top of the main window and click on the ALEX checkbox. Setting “stroboscopic illumination” in addition to ALEX has no effect. Because of the [rolling shutter](#) of CMOS cameras (including the Hamamatsu Fusion and Flash cameras), the duty cycle in ALEX mode can never be 100%. In ALEX mode, Flash Gordon will automatically ensure that the shutter is only ever open during the “global exposure” period where all lines of the camera are exposing on the same frame number (i.e., no lines are currently being read out). With shorter exposure times, there is relatively little “global exposure” period during which to open the shutter. To compensate, Flash Gordon will

automatically select the number of camera lines so that the target duty cycle is achieved. To maximize the size of the field of view, reduce the duty cycle.

WARNING: mechanical shutters have a maximum operating rate. Shuttering too rapidly can permanently damage the device. Check the datasheet for your shutter for details. The maximum rate is set in the configuration file. **Do this before using advanced imaging modes.**

Movie File Format

Flash Gordon saves image stacks (movies) as monochrome, uncompressed .tif files with the 16-bit pixel data copied from the DCIMG intermediate files produced by the Hamamatsu Hard Disk Recorder library. For each frame, the image data is flipped vertically and/or horizontally according to the configuration and the images from all active cameras are then assembled into a montage. The montage is designed to simulate the layout of an optical splitter. For 4-channel acquisitions, the panel order by wavelength is bottom-left, top-left, top-right, bottom-right (Cy2, Cy3, Cy5, Cy7, respectively). For 3-channel acquisitions, the field corresponding to the inactive camera is filled with zeros. For 2-channel acquisitions, the fields are tiled horizontally left-to-right by wavelength.

Movie files use the BigTIFF standard, which uses 64-bit offsets to enable file sizes larger than 4 GB: <https://www.awaresystems.be/imaging/tiff/bigtiff.html>. This is required for instruments with sCMOS cameras, which often produce movies that are 30 GB or larger.

The movie file includes one Image File Directory (IFDs) for each frame, all of which are stored at the beginning of the file (offset=16384 bytes). The pixel data is saved in one contiguous block at the end of the file, with each frame being a single stripe without any intervening bytes. This allows the pixel data to be efficiently read from disk.

The movies include many metadata tags, but these two are of particular importance. The pixel size parameter is set for the Flash cameras and a 60x objective and may not be correct for any other setup.

- ExposureTime (33434): interval between camera frames.
- ImageDescription (270): plain-text description of the experiment and instrument parameters, which can be used to automate the analysis process.

Adding support for new hardware devices (source code version)

This section provides some useful guidelines for adding support for new devices with the source code version of Flash Gordon. It is assumed that the user is already experienced with [LabVIEW programming paradigms](#) and the [Actor framework](#).

DAQ: most National Instruments DAQ devices are likely supported; only the NI-DAQmx physical channel addresses in the configuration file must be changed. If any code changes are needed, they will be in "Hardware\Synchronization Device\".

Shutters: any shutter driver that can be triggered by TTL (0/+5V) logic should be supported by Flash Gordon. Other optical modulators, such as AOM, EOM, and AOTF devices, can be similarly controlled if they can be programmed to accept TTL signals from the DAQ.

Lasers: support for controlling laser power directly at the laser controller or via some attenuator device (such as a motorized rotating polarizer) can be added by modifying the following files in “Hardware\Laser Quantum\”. Each of these functions includes a case structure with a case for each supported device type.

- **Init Device.vi:** establish a connection with the device and perform any startup tasks. For RS232 (serial) connections, this is where baud rate and other parameters are set.
- **Set Laser Power.vi:** send commands to the device to adjust laser power.
- **Shut Down Device.vi:** send any shut down commands and close the connection.

Stage: to add support for a new stage, modify files in “Hardware\Stage Controller\”. Each of these has a Case Structure for each of the supported stages. Be aware that some stages use the same API as the microscope stand, so there may be overlap (see below).

Microscope: to add support for a new microscope stand, and specifically the capability to change lightpath direction and filter set, modify the following files.

- **Resources\Type Defs\Microscope Type.ctl:** add a value for the new API.
- **Hardware\Virtual Microscope\Init Device.vi:** launch a new Actor that you will create. See “Hardware\Ti2\Ti2.lvclass” for an example Actor that you can copy.
- **Hardware\Virtual Microscope\Set Detection.vi:** send messages to the new Actor to direct the microscope to change the lightpath and filter positions.
- **Config file:** modify the Virtual Camera Configuration’s Detection Settings array to include all detection modes you wish to allow in the Main Window. The integer values are meant to be generic and will depend on the specific API.

Cameras: To add support for Hamamatsu cameras, you will need to modify the following files:

- **Resources\Helper VIs\Max Lines.vi:** This function calculates the optimal readout mode and maximum number of camera lines that can be read at the target frame integration time. Currently this assumes 2x2 binning and USB connection, which simplifies the calculation. To add support for a new camera, add a new case to the Case Structure corresponding to the camera model string for the new camera. Read the camera manual to determine the parameters and verify the result with an oscilloscope. If the parameters are not set correctly, the likely result is skipped frames.
- **Hardware\Orca-Flash 4.0 V2\Prepare Acquisition.vi:** This function might need modifications if some standard features are not supported.
- **NOTE:** the Hamamatsu Lightning (C14120-20P) has unusual timing properties that make it harder to support and may require more significant adjustments to the architecture.

Cameras: Adding support for cameras from vendors other than Hamamatsu is possible with the creation of a new actor modeled after the Hamamatsu actor and should inherit the Camera class. Next, the “Camera Type” typedef would need to include a new value for this vendor/camera type/API. Virtual Camera Configuration (xml file line) needs to be changed to the new Camera Type value. Case structures within the Virtual Camera actor class would then need to be modified to utilize the new actor. Finally, new case(s) must be added to “Max Lines.vi” for each supported camera model.

Triggering other devices: to trigger other devices during acquisition using TTL logic (0/+5V), such as fluidics systems or optical modulators (EOM, AOM, AOTF), modify the code in “Hardware\Synchronization Device\”, particularly sync_alex_iteration.vi.

Configuration files: any changes to typedefs will cause problems when loading configuration XML files saved with another definition of the typedef. This is because LabVIEW’s XML schema includes a full definition of the typedef and a list of all possible values in any enum. To avoid this problem, use “Create Hardware Configuration.vi” or modify the files in a text editor.

Troubleshooting

How do I determine the virtual addresses of the ports on the DAQ?

Check the documentation for your DAQ and use the NI MAX utility. NI DAQs often include helpful stickers. If in doubt, use the NI Test Panels utility (accessible from NI Device Manager) to set the line voltages and test with an oscilloscope or volt meter.

Missing DLLs

Flash Gordon works with a set of drivers for common microscope hardware. For licensing reasons, not all of these can be packaged together with the software and must be installed by the user. Check the installation instructions to verify all required software is installed. If the problem persists:

- Ti2_Mic_Driver.dll: install Nikon Ti2 SDK. Then copy this file from “C:\Program Files\Nikon\Ti2-SDK\bin\” into Flash Gordon’s dll folder.
- tmcamcon.dll: install the Hamamatsu Video Capture library.
- dcampapi.dll or dcimgapi.dll: install DCAM (camera drivers).
- Thorlabs....dll: install the Thorlabs Kinesis library.
- nilvaiu.dll: install the National Instruments DAQmx software.

Error -1073807202: A Code Library Required by NI-VISA Could Not Be Located

Although it should have come with DAQmx, you may need to install the NI-VISA driver.

<https://www.ni.com/en-us/support/downloads/drivers/download.ni-visa.html#306119>

Configuration dialog never appears

Check your “Initi.xml” file. If it is invalid, Flash Gordon will close without showing the Configuration Selection dialog. Replace it with the copy that came with the software and try again.

Failed load to configuration

Your configuration file is invalid. We recommend against manually editing these files as this is highly error prone. Start with one of the configuration files provided with Flash Gordon and use the Edit button of the Select Configuration startup dialog to modify the configuration to match your instrument. For the source code version, you can also use the “Create Hardware Configuration.vi” utility.

Microscope is not detected (Nikon Ti2)

You must install the Nikon Ti2 SDK in order for the microscope to be detected. Verify the USB connection and the device shows up in Windows Device Manager.

“No cameras detected!” error on startup. (Hamamatsu cameras)

Verify the cameras are turned on and have completed their startup procedure (status lights not flashing) before starting Flash Gordon. If no lights come on at all, check the power connection. With the Fusion camera, the red lights near the coaxial data ports never stop flashing, it suggests the camera could not establish a connection to the computer; check your USB or coaxial data connection. Finally, double check the configuration file to make sure the serial numbers exactly match. The DCAM Configurator tool can be helpful for this task.

In some cases, the cameras take a long time to initialize and Flash Gordon times out. It’s worth trying a few times before giving up.

If you are using the compiled version but also have LabVIEW and the Hamamatsu Video Capture (HVC) library installed, there may be a conflict with the copy of tmcamcon.dll included with Flash Gordon and the version installed in C:\Windows\System32. To resolve this problem, remove the copy of tmcamcon.dll from the Flash Gordon directory.

If all else fails, use software provided with the camera manufacturer such as HCLImage and work with the camera vendor to troubleshoot the issue.

“Invalid camera model”

Only a limited number of camera models are currently supported by Flash Gordon. Please check the “Supported Hardware” section above. For the source code version, a new model number can be added to the case structure in the “Resources/Max Lines.vi” file. For the stand-alone version, please contact us with the model number.

One of my cameras has an inverted or rotated image relative to the others

You may need to change the Configuration settings for this camera to set one of the “flip” settings to “true”. If it is rotated, this must be fixed by physically rotating the camera so that it is aligned with the others.

I get errors about the lasers

Flash Gordon will not start if any hardware in the configuration file is not found or connecting to it failed. Either change the device type to “Disabled” or delete the Laser Configuration entry corresponding to the device.

Lasers are on and shutters are open but I see nothing in the cameras/eyepieces

First, manually select filter and output port to eyepieces on the microscope body or remote control pad and verify you get the expected signal. If this does not work, the problem may be with the instrument or sample. Otherwise, check your configuration file under Microscope Configuration – Detection Settings. Microscope Type should be “Ti2”. The integer values correspond to the filter turret position (same as written on the filter cubes themselves) and light path (output port). For the Nikon Ti2 microscope stand, the light path has the following values:

- 1) Eyepieces
- 2) Right port
- 3) Bottom (U) port
- 4) Left port

If you see all black in the camera Live Viewer, first adjust the scale bars (or use the Auto Scale mode). This should be showing a rapidly changing static signal – zoom in to verify if this isn’t clear. If the image is completely black or completely static, there may be a problem either with the cameras or with the triggering. The cameras will not acquire frame data unless they receive triggering pulses on their “Trigger Input” line from the DAQ. Check the physical connections and verify the output port specified in the configuration file under Virtual Camera – Sync Line is connected to the “external trigger” port of the camera.

When I move to a new field, some of the molecules near the border are already bleached

Adjust the stage parameters in the configuration file to take a larger step in that direction. These values will depend on the size of the illuminated area.

A movie completed streaming, but was never saved. Can I rescue my data?

You may be able to recover the movie from the .dcimg files. In the main window, select Help -> Rescue Raw Data. Input all relevant information, click “RESCUE”, and you will be prompted to save the resulting .tif file. With the source code version, you can also close shut down Flash Gordon and open the file “Binary TIFF/Class/Rescue Raw Data.vi” from within the project. Input all acquisition parameters manually and click the run arrow in the toolbar.

If this process fails, your movie is unrecoverable.

Flash Gordon is in a strange state where buttons do not work or behave unexpectedly

Stand-alone version: close Flash Gordon. Verify the icon in the taskbar shows that it successfully exited. If it does not, use Task Manager to kill the process. Turn off the cameras, wait 10 seconds, and turn them on again. Wait for the cameras to complete their startup procedure and start Flash Gordon. If this does not solve the problem, restart your computer.

Source code version: close Flash Gordon and wait 1 minute for it to exit completely, then close the project file. If after 1 minute LabVIEW is still running, open Task Manager and kill LabVIEW

completely. Turn off the cameras, wait 10 seconds, then turn them on again. Wait for the cameras to complete their startup procedure (lights not flashing) and run Flash Gordon again. If this does not work, restart the computer.

Still need help?

Please email scott.blanchard@stjude.org with the following information:

1. Name, institution, principle investigator.
2. Version of Flash Gordon and type of installation (stand-alone or source code).
3. Describe the problem in detail. Take a screenshot of any error dialogs.
4. Is the problem reproducible? If so, what are the steps to reproduce it?
5. If using the source code, have there been any modifications? If so, have you tried using the original version?
6. Describe your instrument in detail:
 - a. Camera make, model, and connection interface (e.g., USB). Are you using dedicated PCIe cards for each camera?
 - b. Microscope stand and stage model and connection interface.
 - c. DAQ make and model. If possible, use an oscilloscope to investigate whether Flash Gordon is sending the correct signals to devices as expected.
 - d. Please also include a copy of your configuration file.
7. Describe your computer and installed software:
 - a. Windows version and computer hardware: In Windows Control Panel, select "About your PC". Take a screenshot to include with your report.
 - b. What version of the Hamamatsu DCAM software is installed?
You can find this information in Windows Control Panel under "Add/Remove Programs" in Windows Control Panel or by running DCAM Configurator utility. If there are multiple versions installed, please include this information.
 - c. Source code version: what version of LabVIEW and HVC are you using?
 - d. Versions of all other relevant software: NI DAQmx, Nikon Ti2 SDK, etc.

Known Issues

None